A TRACEABLE DETERMINISTIC DECISION SYSTEM FOR CURRICULUM-BASED ELIGIBILITY ASSESSMENT

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ABSTRACT

This study develops and evaluates a traceable deterministic decision system for curriculum-based eligibility assessment in a virtual-university information system. The problem addressed is the need to transform documented curricular requirements into an executable decision rule while keeping rule construction separate from subsequent evaluation against reference labels. The system represents semester-status conditions and an ECTS threshold as an explicit deterministic predicate, validates and locks the specification before execution, and records the resulting decision for each case. Evaluation was conducted on a controlled corpus of 100 cases. The system produced 20 ELIGIBLE and 80 NOT_ELIGIBLE decisions, with TN=80, FP=0, FN=0, and TP=20. Accuracy, Precision, Recall, F1, Specificity, Balanced Accuracy, and MCC were all 1.000000, with a 95% confidence interval of [0.963007, 1.000000] for Accuracy. The results establish agreement for the evaluated corpus and protocol, not generalization to other curricula or cohorts.

KEYWORDS

Academic decision support, curriculum-based eligibility, decision traceability, deterministic decision system, higher education information systems

1. INTRODUCTION

Eligibility assessment is a recurring operation in higher-education information systems. A decision such as whether a student satisfies a curriculum-based eligibility condition requires the system to apply several documented academic requirements to a structured student record and produce a reproducible outcome. When these requirements are implemented as implicit or poorly documented processing steps, it becomes difficult to determine which conditions produced a particular decision and to reconstruct the decision for audit or verification.

The problem addressed in this study is therefore not prediction of student success or recommendation of academic choices [1], [2], [3]. It is the formal and traceable implementation of a predefined curriculum-based eligibility criterion[4]. The specific challenge is to transform documented curricular requirements into an executable deterministic rule, validate the rule before evaluation, preserve the provenance of the decision inputs and conditions, and subsequently

compare the resulting decisions with reference labels without using those labels to construct the rule.

This study does not employ machine learning to infer the eligibility boundary. The decision boundary is specified from documented curricular requirements before evaluation. Consequently, the contribution concerns deterministic decision-system engineering, rule execution, traceability and empirical validation rather than model training or predictive learning. Machine-learning and recommender-system studies are considered as related work because they address broader academic decision-support problems, but their predictive performance is not used as a direct benchmark for the present deterministic artifact[5]–[7].

The study follows a design-science orientation in which an information-system artifact is developed and evaluated against a defined problem context [8], [9]. The contribution is deliberately narrower than a general academic advising architecture: it formalizes a curriculum-derived decision mechanism so that its conditions, inputs and outputs are explicit, deterministic and traceable. The study therefore contributes a decision artifact and an evaluation protocol, rather than a learned predictive model or a post-hoc explanation method.

The specific research gap is the need for a clearly separated workflow that maps a fixed curriculum specification to an executable eligibility predicate, validates that predicate independently of evaluation labels, and preserves traceability from specification to empirical evidence. The proposed approach addresses this gap through three linked elements: formalization of the decision predicate, separation of construction and execution from label-based evaluation, and explicit provenance of the resulting evidence.

The empirical claim is intentionally restricted to the specified rule, controlled corpus and evaluation protocol. Perfect concordance is therefore interpreted as agreement with the reference labels for the evaluated cases, rather than as evidence of universal correctness or transferability.

2. RELATED WORK

2.1. Rule-Based and Deterministic Decision Support

Rule-based decision support has been applied to recurring academic decisions in higher education. Engin et al. reported university expert systems for course advising and scholarship eligibility, illustrating how explicit rules can operationalize eligibility criteria and support decisions that would otherwise require substantial manual advising. This establishes the relevance of rule-based decision support to academic eligibility contexts. The present work extends that line of inquiry by treating the eligibility decision as a deterministic and traceable decision artifact, in which the governing curricular conditions and the resulting decision can be inspected together. [10]

2.2. Traceability, Provenance and Auditability

Traceability and provenance provide a complementary perspective to rule-based decision support by making the evidentiary path of an automated decision inspectable. Recent work on automated decision provenance argues that provenance can serve as a foundation for explaining how a decision was produced, while recent audit-oriented frameworks operationalize traceability through versioned artefacts, lineage metadata, audit trails, and reproducibility records. [11], [12]. For the present problem, this distinction is important: recording that a decision occurred is weaker than preserving the decision context, the applicable rules or criteria, and the evidence

required to reconstruct the decision. The proposed system therefore treats traceability as a property of the decision artefact itself rather than as a separate post-hoc reporting layer. This provides a direct bridge between deterministic eligibility rules and auditable academic decisions.

2.3. Curriculum-Based Eligibility and Academic Decision Support

Curriculum-based academic decision support commonly operationalizes degree requirements by comparing a student's completed or planned coursework with the rules of a program. Degree audit systems implement this principle by evaluating requirements such as required courses, credits, course sequences, minimum grades, catalog-year rules, and other conditions. The literature also shows that such systems are used as advising tools rather than as unqualified replacements for academic judgment. [13], [14]. Chénier and White demonstrated that academic audit requirements can be represented through conjunctions and disjunctions of complex terms, including temporal course conditions and changing credit values. [15]. More recent curriculum-analysis and advising research extends this paradigm toward curriculum modeling and personalized academic planning, but still emphasizes the role of human interpretation in non-standard cases. [16]. These studies establish the relevance of curriculum-aware decision support while leaving a distinct methodological space for a deterministic eligibility mechanism in which the applicable curriculum conditions and the resulting decision remain explicitly inspectable and reproducible.Conventional degree-audit systems and the present artifact share the use of explicit curricular requirements to evaluate academic status. The distinction addressed here is not that the proposed rule replaces degree auditing, but that the study isolates one eligibility predicate as a separately specified and evaluated decision artifact. The evaluation therefore focuses on the provenance of the decision boundary, the separation between rule construction and reference-label evaluation, and the reconstruction of individual decisions. These properties are examined independently of the broader advising and degree-progress functions normally provided by degree-audit systems.

2.4. Research Gap and Distinction

The relevant distinction is methodological rather than merely application-specific. Existing academic decision-support systems address broader tasks such as advising, recommendation and prediction [17], [1], [2]. The present work isolates a predefined eligibility decision and makes its decision boundary executable and inspectable. This creates a different evaluation object: the correspondence between a documented criterion, its implementation and the resulting case-level decisions.

A predefined eligibility predicate also differs from a learned decision function. In the former, the conditions are specified before evaluation and changes to those conditions constitute specification changes; in the latter, the decision boundary is inferred from training data. This distinction is central to the methodology because ground-truth labels remain evaluation targets rather than inputs to rule construction.

Accordingly, comparison with related systems is made using decision objective, origin of the decision boundary and inspectability of the decision conditions. Agreement values from unrelated predictive or recommender tasks are not treated as directly comparable performance measures.

The contribution of this study does not reside in the complexity of the eligibility predicate itself. The predicate is intentionally simple because it represents a documented curricular condition. The contribution lies in the engineering and evaluation procedure used to make this condition an explicit and auditable decision artifact. More specifically, the proposed approach establishes a separation between curriculum-derived rule construction and label-based evaluation; requires

operational validation and specification locking before execution; preserves the provenance of the decision inputs and evaluated conditions; and produces case-level decision records that can be reconstructed from the fixed specification. The empirical evaluation then measures the correspondence between these independently executed decisions and the reference labels. The resulting contribution is therefore a traceable decision-system methodology for a predefined academic eligibility criterion, rather than a novel predictive algorithm.

3. FORMALIZATION AND METHODOLOGY

3.1. Scientific Contribution and Design Rationale

The scientific contribution is organized around three properties of the decision artifact. First, the curricular criterion is formalized as an explicit executable predicate rather than remaining an implicit administrative procedure. Second, rule construction, operational validation and empirical evaluation are separated so that reference labels do not participate in defining the decision boundary. Third, the resulting decisions are associated with provenance and execution information so that the relationship between curricular requirements, evaluated conditions and final decisions can be inspected. The contribution is consequently methodological and architectural: it defines how a simple curriculum rule can be transformed into a controlled, versioned and auditable decision component and then evaluated under a fixed protocol.

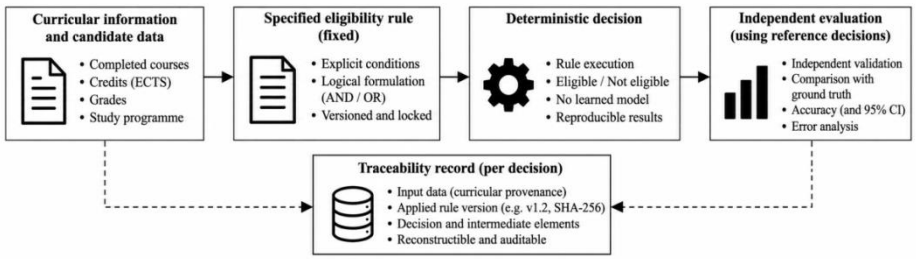


Figure 1. Conceptual model of traceable deterministic decision assessment.

3.2. Formal Decision Model

The artifact is defined by three properties: explicitness, determinism and traceability. Explicitness means that decision conditions are represented by observable input variables. Determinism means that a fixed specification and fixed input record yield one reproducible binary outcome. Traceability means that the outcome can be connected to the specification, input record and evaluation record. These properties characterize the decision mechanism itself and are independent of its later empirical performance. [18]

Determinism is therefore an operational property, not a statistical performance measure. For fixed inputs and a fixed specification, repeated execution is expected to return the same outcome. Accuracy and related measures are evaluated only after execution against the reference labels.
The formal decision function is given in Eq. (1).

Let S_i denote that semester i has VALID status, and let E denote ECTS_OBTAINED.

$$D(x) = ELIGIBLE \Leftrightarrow (S^1 \wedge S^2 \wedge S^3 \wedge S^4 \wedge E \geq 120) \quad (1)$$

Otherwise, D(x) = NOT_ELIGIBLE.

The system determines the ELIGIBLE versus NOT_ELIGIBLE decision from the semester-level status variables and the accumulated European Credit Transfer and Accumulation System (ECTS) value specified by the locked decision system.

3.3. Traceability and Provenance Model

Curriculum provenance for S1, S2, S3 and S4 was established from the corresponding curriculum documents. The provenance evidence establishes the curricular basis of the operational inputs and ECTS structure; it is distinct from the computation rule itself.

3.4. Rule-Based and Deterministic Decision Support

After operational validation, the decision specification was finalized and locked before evaluation. For a case x satisfying the conditions in Eq. (1), the implemented decision is D(x)=ELIGIBLE; otherwise, D(x)=NOT_ELIGIBLE. Thus, Eq. (1) defines the operational decision boundary used in the evaluation.

Because the boundary is explicit, changing a semester-status condition or the ECTS threshold changes the decision specification rather than tuning a learned model. The specification is consequently treated as a versioned artifact whose identity remains fixed during evaluation.

The specification also records UE and semester validation, the 30-credit semester principle, and compensation as an authorized mechanism. The compensation threshold and implementation, the DEFECT_CODE role, and the exclusion condition remained explicitly bounded or unresolved in the locked specification. S4_STATUS is treated as an input status variable in the executed rule; the internal procedure used to derive this status is outside the scope of the present evaluation. These elements were not silently replaced during execution; consequently, claims about these unresolved operational elements are outside the empirical scope of the present evaluation.

The main elements of the locked specification are summarized in Table 1.

Before execution, the candidate rule was checked for schema compatibility, logical consistency, determinism and the ECTS boundary condition. Ground-truth labels were excluded from this validation stage. This ordering prevents evaluation targets from influencing construction or operational validation of the decision rule.

The dataset SHA-256 digest was verified before execution, providing an integrity check on the corpus used for the reported evaluation.

The locked decision rule was applied to the 100 cases, producing one decision record per case. Ground truth was not used in decision construction and was introduced only for evaluation. The resulting execution record was preserved as the empirical basis for the reported comparison.

Table 1. Formal specification of the eligibility rule.

Element	Definition	Role
Input record	Curriculum / candidate information evaluated by the system	Decision input
Eligibility conditions	Predicates specified by the decision rule	Condition evaluation

Decision rule	Deterministic evaluation of the specified conditions	Decision generation
Decision	ELIGIBLE / NOT ELIGIBLE	Final outcome
Traceability	Link between input, evaluated conditions and decision	Auditability

4. EVALUATION METHODOLOGY

4.1. Evaluation Sequence

The evaluation comprised three predefined phases: independent validation of the operational rule; execution of the locked rule on the controlled corpus; and comparison of the resulting decisions with the reference labels using the predefined measures. This sequence preserves the separation between decision construction and empirical assessment [19].

The completed workflow was documented through traceability and critical-audit artifacts. These artifacts preserve the provenance of the experimental record and support reconstruction of the documented procedure without introducing new empirical analysis.

Because the rule conditions are explicit, an individual decision can be inspected through the predicates that determine the outcome. This is a property of the decision artifact itself and differs from explanation methods that infer or approximate the rationale of a learned model [20], [21]. It also supports direct inspection of the decision record rather than post-hoc auditing [22]. Requirements traceability literature likewise emphasizes explicit links between requirements and downstream artefacts [23].

Post-hoc explanation methods such as LIME address interpretation of predictions produced by black-box classifiers [24]. In the present setting, the decision conditions are specified as part of the artifact before execution, so inspection does not require a separate explanatory model.

4.2. Ground-Truth Construction and Independence

The reference labels constitute the evaluation target and were not used during rule construction or operational validation. The labeling procedure, source records, decision criteria applied during labeling, and sequence of label generation must be documented separately from the implementation of the deterministic rule. This separation is required to distinguish agreement with an independently established reference from agreement resulting from reuse of the same rule. The 100 reference labels were therefore treated as fixed evaluation data throughout execution.

4.3. Case-Level Decision Trace

To illustrate the traceability mechanism, two representative decision records are presented: one ELIGIBLE case and one NOT_ELIGIBLE case. For each case, the trace contains the input status variables, the ECTS value, the evaluated predicates, the resulting decision, and the corresponding audit information. The examples are intended to demonstrate reconstruction of an individual decision from the locked specification rather than to constitute additional evaluation cases.

5. RESULTS

The evaluation corpus contained 100 cases. The deterministic rule classified 20 cases as ELIGIBLE and 80 as NOT_ELIGIBLE. Comparison with the reference labels produced 80 true negatives, 20 true positives, zero false positives and zero false negatives. Consequently, Accuracy, Precision, Recall, F1, Specificity, Balanced Accuracy and MCC were all 1.000000. The reported 95% confidence interval for Accuracy was [0.963007, 1.000000]. Thus, every evaluated case received the same class as its reference label. These results quantify agreement within the tested corpus; they do not establish independent validity unless the reference-label construction is shown to be independent of the rule.

Table 2. Confusion matrix for the controlled evaluation corpus.

Actual / Predicted	NOT_ELIGIBLE	ELIGIBLE
NOT_ELIGIBLE	TN = 80	FP = 0
ELIGIBLE	FN = 0	TP = 20

Table 3. Locked evaluation results.

Metric	Value
TN	80
FP	0
FN	0
TP	20
Accuracy	1.000000
Precision	1.000000
Recall	1.000000
F1	1.000000
Specificity	1.000000
Balanced Accuracy	1.000000
MCC	1.000000
95% CI for Accuracy	[0.963007, 1.000000]

6. DISCUSSION

Comparative interpretation. Academic advising and decision-support literature addresses computational assistance for planning, course selection, advising and related decisions [17], [1], [2], [3]. The present artifact addresses a narrower task: executing an explicitly specified curriculum-derived eligibility predicate. The comparison is therefore based on decision objective and mechanism, not on a claim of superiority.

Recommendation-oriented systems may rank alternatives or personalize choices [17], [1], while predictive and early-warning systems estimate outcomes or identify students requiring intervention [4]. The present artifact instead evaluates a predefined eligibility condition. These objectives imply different target variables and evaluation questions, so reported performance values from those systems are not treated as directly comparable to the present agreement measures.

The principal methodological distinction is the origin of the decision boundary. Here, the boundary is specified from curricular requirements before evaluation, and the resulting decisions

can be traced to explicit predicates. This provides inspectability of the rule and reproducibility of its execution without requiring a learned predictive model.

Under the tested corpus and locked protocol, the observed concordance demonstrates correspondence between the operational rule and the reference labels represented in that corpus. The evidential scope is therefore the specified rule, corpus and protocol.
The observed agreement supports the bounded empirical claim that the implemented rule reproduced the reference labels for all evaluated cases. It does not establish universal correctness or performance under a different curriculum, cohort or ground-truth construction.

Several operational elements remain explicitly bounded in the locked specification, including the compensation threshold, compensation implementation, derivation of S4_STATUS, DEFECT_CODE role and exclusion condition. These boundaries define the current scope of the decision artifact. Resolving any of them would require a new controlled specification version with preserved traceability.

The perfect agreement observed in the 100-case corpus should be interpreted in relation to the construction of the reference labels. If the reference labels were independently established from the underlying academic records, the result provides evidence that the implemented rule reproduced those labels for the evaluated cases. If the labels were derived from the same curricular rule or from a process that directly reused its decision conditions, the result instead demonstrates internal consistency between the implementation and its reference construction. The distinction is essential because identical outputs do not by themselves establish independent validation.

7. CONCLUSIONS

The study developed and evaluated a traceable deterministic decision artifact for a predefined curriculum-based eligibility criterion. On the controlled corpus of 100 cases, the system produced 20 ELIGIBLE and 80 NOT_ELIGIBLE decisions and obtained 80 true negatives, 20 true positives, zero false positives and zero false negatives. The resulting evaluation metrics were all 1.000000. These results establish exact agreement between system outputs and the reference labels for the evaluated cases, subject to the independence and construction procedure of those labels.

The findings support the specific claim that, under the locked specification and tested protocol, the implemented rule produced decisions consistent with the evaluated labels. The contribution is the traceable operationalization of a predefined eligibility criterion, not evidence of superiority over systems addressing different decision objectives.

The study establishes an evidence chain from curriculum-derived specification to executable rule, decision record and empirical evaluation. Validation on additional curricula or cohorts would constitute a distinct empirical extension of this decision artifact.

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